

+20 12 7247 3302
Cairo, Egypt
hamzahassanain067@gmail.com

Hamza Mohammed Hassanain

Software Engineer

github.com/HamzaHassanain
gitlab.com/HamzaHassanain
linkedin.com/in/hamza-hassanain-ha067
dev.to/hamzahassanain0

SUMMARY

Founding Software Engineer at **Repovive** (YC F'26), where I built the contest judge and core infrastructure behind **10,000+ users**. **AWS Certified Solutions Architect – Associate** and **3× ACPC finalist**. I contribute to open source (GitLab, LLVM, Swift NIO) and write about systems and memory on dev.to.

EDUCATION & CERTIFICATIONS

B.Sc. Computer and Information Sciences, Suez Canal University
AWS Certified Solutions Architect – Associate, Amazon Web Services

2022 – 2026
credly.com/verify

EXPERIENCE

Founding Software Engineer
Repovive (YC F'26)

Nov 2025 – Present
Full Time – United States, Remote

Repovive is a Y Combinator-backed science expert community where experts solve unsolved scientific problems, starting with theoretical computer science and math proof reasoning. We host live programming and math contests and have grown to 10,000+ users.

- Joined as the **first engineer** and built the core systems from the early days through YC.
- Onboarded and supported new engineers, senior hires included, guiding them through the codebase and projects.
- Led design of the distributed judge (judge1 on **Judge0**); scaled submissions **10x (50 to 500+ concurrent)**.
- Cut infrastructure cost **95% (\$600 to \$25/mo)** by migrating resource-heavy tasks from serverless to Docker/Nginx/Redis.
- Isolated **20+ languages** in resource-limited Docker images; redesigned the DB layer for **50MB+** contest test cases.
- Collaborated with the team to design and build AI voice interviews (**LiveKit**): recruiter, behavioral, and coding agents.
- Moved CI to **Vercel** when Actions quota ran out: one fail-fast pipeline (format, typecheck, test, build), no prod secrets.
- Tracked down live-contest outages with **Claude Code**, **MCP servers**, **PostHog**, and log analysis, then shipped the fixes.

OPEN SOURCE CONTRIBUTIONS

- GitLab Go SDK (client-go)**: typed group integrations – Slack, Jira, MS Teams, Mattermost, Harbor, +3 | [!2692](#), [!2691](#), [!2675](#), [!2670](#).
- GitLab Terraform provider**: new Harbor & Mattermost integration resources + group SAML attribute | [!2881](#), [!2909](#), [!2852](#).
- GitLab monorepo**: `Json.safe_parse` hardening and Vue lint compliance | [!221163](#), [!221158](#).
- LLVM / Clang**: `constexpr X86 cvtpd2ps`, `AArch64/ARM CLS` → `ISD`: `:CTLS`, `LLDB SWIG` tests | [#169980](#), [#178885](#), [#178430](#), [#173473](#).
- Apple Swift NIO**: idempotent `NIOFileSystem` directory creation | [#3410](#).

PROJECTS

Autonomous AI Tutor & Curriculum Engine, Graduation Project

github.com/sigma-loop

- Led backend & DevOps for an 8-person team building an on-demand AI course, lesson, and challenge generator.
- Graded code in a Judge0 sandbox and math proofs via a confidence-scored LLM; owned all CI/CD and AWS deployments.

Competitive-Programming Judge — AWS Reference Architecture

github.com/HamzaHassanain/CP-Judge-AWS

- Designed an AWS reference architecture: Judge0 `isolate` on ECS-on-EC2, Fargate orchestrator, Spot ASG on queue depth.
- Automated post-round cheating detection with Step Functions, AWS Batch, Lambda/Bedrock, and Athena.

Polyman CLI

npmjs.com/package/polyman-cli | github.com/HamzaHassanain/polyman

- Built a TypeScript CLI automating the Codeforces Polygon workflow: test generation, validation, and direct API sync.

Systems in C++ — HALlocator & cppress

[HALlocator](#) | [cppress](#)

- Built a Red-Black-Tree best-fit memory allocator ($O(\log n)$) and an epoll-driven C++ web framework, both from scratch.

TECHNICAL WRITING

- Wrote ["What Every Programmer Should Know About Memory"](#) and ["I Built a Backend Web Framework from Scratch in C++"](#).

ACHIEVEMENTS

ICPC ACPC — 3× Finalist (2024, 2025, 2026)

icpc.global/YC7Z02WQCPIB

17th of 500 teams at ECPC (2024); top 250 at ACPC across the Africa & Arab region.

Problem Author — The 2025 ICPC Jordanian Collegiate Programming Contest (JCPC)

Wrote [G. Queries On Strings](#) for JCPC 2025, and [F. Graph Tiling](#) for Repovive Standard Round 3.

SKILLS

Cloud & DevOps

AWS (Certified Solutions Architect), Docker, Nginx, Redis, Terraform, CI/CD, DigitalOcean, Linux

Backend & Full-Stack

TypeScript, Node.js, Express.js, Next.js, React, MongoDB, PostgreSQL, REST APIs, WebSockets, Yjs (CRDTs)

Languages & Systems

C++, Go, Rust, Python, SQL; System Design, Distributed Systems, Message Queues